Functions & Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating functions**

Consider $f(x) = 2x^2 - 5x + 1$. Evaluate:

a $f(0)$ 1

b $f(2)$ -1

(a) 1; (b) -1

2 **[Foundational]** • **Target: Domain and range**

The graph of $y = \sqrt{x + 3}$ is shown below. State the domain and range based on the graph, then verify the domain algebraically.

From the graph, the domain is $x \geq -3$ and the range is $y \geq 0$. Algebraically, the expression under the root must be non-negative: $x + 3 \geq 0$, so $x \geq -3$, confirming the graph.

3 **[Foundational]** • **Target: Domain and range**

Find the domain of $f(x) = \frac{1}{x-5}$.

The denominator cannot equal zero, so $x \neq 5$. Domain: all real numbers except 5.

4 **[Foundational]** • **Target: Composition of functions**

If $f(x) = x + 5$ and $g(x) = 3x$, find $(g \circ f)(x)$ as a simplified expression.

$g(f(x)) = 3(x + 5) = 3x + 15$.

5 **[Foundational]** • **Target: Composition of functions**

If $f(x) = x^2$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and state whether they are equal.

$f(g(x)) = (x - 4)^2 = x^2 - 8x + 16$. $g(f(x)) = x^2 - 4$. These are not equal, so function composition is generally not commutative.

6 **[Foundational]** • **Target: Inverse functions**

Find the inverse of $f(x) = 4x - 8$.

Swap x and y in $y = 4x - 8$: $x = 4y - 8$, so $y = \frac{x+8}{4}$. $f^{-1}(x) = \frac{x+8}{4}$.

7 **[Foundational]** • **Target: Inverse functions**

Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses of each other by computing $f(g(x))$.

$f(g(x)) = 5\left(\frac{x-1}{5}\right) + 1 = (x - 1) + 1 = x$, confirming they are inverses.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Transformations of graphs**

The graph of $y = f(x)$ is shifted 4 units left and 2 units down. Write the new function in terms of f.

$y = f(x + 4) - 2$.

9 **[Intermediate]** • **Target: Transformations of graphs**

If $f(x) = x^2$ and $g(x) = -f(x) + 3$, describe the transformation from f to g.

g reflects f over the x-axis, then shifts the result up 3 units.

10 **[Intermediate]** • **Target: Piecewise functions**

A function is defined by $f(x) = x + 2$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Find $f(-3)$ and $f(3)$.

$f(-3) = -3 + 2 = -1$ (using the first piece, since $-3 < 0$). $f(3) = 3^2 = 9$ (using the second piece, since $3 \geq 0$).

11 **[Intermediate]** • **Target: Piecewise functions**

A function is defined by $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(2)$ and $f(5)$, and state whether f is continuous at $x = 2$.

$f(2) = 2(2) = 4$ (using the first piece, since $2 \leq 2$). $f(5) = 5 + 5 = 10$. As $x \to 2$ from the right, $f(x) \to 2 + 5 = 7$, which does not match $f(2) = 4$, so f is not continuous at $x = 2$ — there is a jump.

12 **[Intermediate]** • **Target: Reading function graphs**

The graph of $f(x) = x^2 - 4$ is shown below. Use the graph to estimate the x-intercepts and y-intercept, then find them exactly.

x-intercepts: solve $x^2 - 4 = 0$, giving $x = 2$ and $x = -2$. y-intercept: $f(0) = -4$.

13 **[Intermediate]** • **Target: Even, odd, and other function properties**

Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, using $f(-x)$.

$f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$, so f is even.

14 **[Intermediate]** • **Target: Even, odd, and other function properties**

Determine whether $f(x) = x^3 - x$ is even, odd, or neither.

$f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$, so f is odd.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: A well-designed word problem: modeling a car rental cost** • **Cross-Lesson: 4.01 Ratios, Rates & Data**

This question has three parts. A rental car company models the cost of a rental as $C(d) = 45d + 20$ dollars, where d is the number of days rented. (a) Find $C(5)$, and interpret its meaning in context. (b) Define $P(d) = C(d) - 200$, representing profit relative to a fixed cost of 200 dollars per rental period. Write $P(d)$ explicitly and find $P(5)$. (c) Find the inverse function $C^{-1}(x)$, and use it to determine how many days someone rented the car if they were charged 335 dollars.

(a) $C(5) = 45(5) + 20 = 245$. Renting the car for 5 days costs 245 dollars total. (b) $P(d) = 45d + 20 - 200 = 45d - 180$. $P(5) = 45(5) - 180 = 225 - 180 = 45$ dollars. (c) From $x = 45d + 20$, solve for d: $d = \frac{x-20}{45}$, so $C^{-1}(x) = \frac{x-20}{45}$. For $x = 335$: $d = \frac{335-20}{45} = \frac{315}{45} = 7$ days.

16 **[Challenging]** • **Target: Reading function graphs** • **Cross-Lesson: 5.01 Statistics & Probability**

Determine the maximal real domain of the function $f(x) = \frac{\sqrt{3x-12}}{x^2-25}$. Express your answer in interval notation.

Step 1: For the square root in the numerator to be defined over the reals, the radicand must be non-negative: $3x - 12 \geq 0 \Rightarrow 3x \geq 12 \Rightarrow x \geq 4$. Step 2: For the denominator to be non-zero: $x^2 - 25 \neq 0 \Rightarrow x \neq 5$ and $x \neq -5$. Step 3: Intersect the conditions $x \geq 4$ and $x \neq 5$ (note that $x = -5$ is already outside $x \geq 4$): $[4, 5) \cup (5, \infty)$. The maximal real domain is $\mathbf{[4, 5) \cup (5, \infty)}$.

17 **[Challenging]** • **Target: Even, odd, and other function properties** • **Cross-Lesson: 6.01 Geometry**

Given $f(x) = x^2 - 4x + 7$ with restricted domain $x \geq 2$, find an explicit formula for $f^{-1}(x)$ and state its domain.

Step 1: Complete the square to isolate x: $y = (x^2 - 4x + 4) + 3 = (x - 2)^2 + 3$. Step 2: Swap x and y (with $y \geq 2$): $x = (y - 2)^2 + 3 \Rightarrow (y - 2)^2 = x - 3$. Step 3: Take the principal square root (since $y \geq 2 \Rightarrow y - 2 \geq 0$): $y - 2 = +\sqrt{x - 3} \Rightarrow y = 2 + \sqrt{x - 3}$. Therefore, $\mathbf{f^{-1}(x) = 2 + \sqrt{x - 3}}$, defined for domain $\mathbf{x \geq 3}$ (which matches the range of f).

18 **[Challenging]** • **Target: A well-designed word problem: modeling a car rental cost** • **Cross-Lesson: 7.01 Trigonometry & Additional Topics**

Let $f(x) = \frac{2x+1}{x-3}$ for $x \neq 3$. (a) Determine the inverse function $f^{-1}(x)$. (b) State the domain and range of $f^{-1}(x)$.

(a) Find the inverse function: Set $y = \frac{2x+1}{x-3}$. Swap x and y: $x = \frac{2y+1}{y-3} \Rightarrow x(y - 3) = 2y + 1 \Rightarrow xy - 3x = 2y + 1$. Group y terms: $xy - 2y = 3x + 1 \Rightarrow y(x - 2) = 3x + 1 \Rightarrow \mathbf{f^{-1}(x) = \frac{3x+1}{x-2}}$. (b) Domain and Range: The denominator of $f^{-1}(x)$ cannot be zero, so the domain is $\mathbf{(-\infty, 2) \cup (2, \infty)}$ (all $x \neq 2$). The range of f^{-1} equals the domain of f, which is.

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19 [Challenging] • Target: Evaluating functions • Cross-Lesson: 8.01 Exponentials & Logarithms

Let $f(x) = \frac{2x-7}{5}$. (a) Find $f^{-1}(x)$. (b) Solve the equation $f(f^{-1}(x)) = 9$. (c) Find the value of k for which $f(k) = f^{-1}(k)$.

(a) Find $f^{-1}(x)$: Let $y = \frac{2x-7}{5}$. Swap x and y: $x = \frac{2y-7}{5} \Rightarrow 5x = 2y - 7 \Rightarrow 2y = 5x + 7 \Rightarrow \mathbf{f^{-1}(x) = \frac{5x+7}{2}}$. (b) Composition identity: By definition, for all x in the domain of f^{-1}, $f(f^{-1}(x)) = x$. Therefore, $f(f^{-1}(x)) = 9 \Rightarrow \mathbf{x = 9}$. (c) Point on line $y = x$: If $f(k) = f^{-1}(k)$, the intersection must lie on the line of reflection $y = x$. Setting $f(k) = k$.

20 [Challenging] • Target: Domain and range • Cross-Lesson: 1.01 Heart of Algebra

The graph of $y = f(x)$ passes through the points $(-2, 8)$ and $(4, -6)$. If $g(x) = 3f(x-5) + 2$, determine the corresponding points on the graph of $g(x)$.

Transformation Mapping: For any point (x, y) on f, the corresponding point on $g(x) = 3f(x-5) + 2$ has coordinates: $(x', y') = (x + 5, 3y + 2)$. For point 1: $(-2, 8)$:

$x'_1 = -2 + 5 = 3, \quad y'_1 = 3(8) + 2 = 24 + 2 = 26 \Rightarrow \mathbf{(3, 26)}$. For point 2: $(4, -6)$:

$x'_2 = 4 + 5 = 9, \quad y'_2 = 3(-6) + 2 = -18 + 2 = -16 \Rightarrow \mathbf{(9, -16)}$. The corresponding points on $g(x)$ are $\mathbf{(3, 26)}$ and $\mathbf{(9, -16)}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating functions

What is the domain of the real-valued function $f(x) = \sqrt{2x - 14}$?

a. $x \geq -7$

b. $x \leq 7$

c. $x > 7$

d. $x \geq 7$ ✓

For the function $f(x) = \sqrt{2x - 14}$ to produce real values, the radicand must be non-negative: $2x - 14 \geq 0$ Add 14 to both sides: $2x \geq 14$ Divide by 2: $x \geq \frac{14}{2} = 7$ Thus, the domain is all real numbers such that $x \geq 7$.

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22 [Foundational] • Target: Domain and range

If $f(x) = \frac{2x+3}{x-4}$ for $xeq4$, what is the inverse function $f^{-1}(x)$?

a. $\frac{4x+3}{x-2}$ ✓

b. $\frac{x-4}{2x+3}$

c. $\frac{4x-3}{x+2}$

d. $\frac{2x-3}{x+4}$

Set $y = \frac{2x+3}{x-4} \Rightarrow y(x-4) = 2x+3 \Rightarrow xy-4y = 2x+3 \Rightarrow xy-2x = 4y+3 \Rightarrow x(y-2) = 4y+3 \Rightarrow x = \frac{4y+3}{y-2}$. Thus $f^{-1}(x) = \frac{4x+3}{x-2}$.

23 [Foundational] • Target: Composition of functions

The graph of the function $y = ax^2 + bx + c$ is shown below. The function passes through the points $(1, 0)$ and $(5, 0)$, and its vertex has a y-coordinate of -4. Determine the value of b.

a. -6

b. **6** ✓

c. -4

d. 4

The roots of the quadratic function are given as $x = 1$ and $x = 5$. Therefore, the quadratic can be written in the form $y = a(x-1)(x-5)$. The vertex of a parabola is located at the midpoint of its roots, so the x-coordinate of the vertex is $x_v = \frac{1+5}{2} = 3$. We are given that the y-coordinate of the vertex is -4. Substituting the vertex coordinates into the factored form of the quadratic: $-4 = a(3-1)(3-5)$ $-4 = a(2)(-2)$ $-4 = -4a$ $a = 1$ Now, substitute $a = 1$ back into the factored form.

24 [Foundational] • Target: Inverse functions

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(1, 0)$ and $(5, 0)$. If the vertex of the parabola is at $(3, -4)$, what is the value of a?

a. 1

b. **-1** ✓

c. 2

d. -2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. Since the parabola has x-intercepts at $(1, 0)$ and $(5, 0)$, we can write the function in factored form: $f(x) = a(x-1)(x-5)$. The x-coordinate of the vertex of a parabola in this form is the midpoint of the x-intercepts, which is $\frac{1+5}{2} = 3$. This matches the given x-coordinate of the vertex. We are also given that the vertex is at $(3, -4)$. We can substitute these coordinates into the factored form of the function to find the value.

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25 [Foundational] • Target: Transformations of graphs

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(-1, 0)$ and $(3, 0)$. The vertex of the parabola is at $(1, -4)$. Determine the value of a.

a. -1

b. 1 ✓

c. 2

d. -2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. Since the parabola has roots at $x = -1$ and $x = 3$, we can write the function in factored form as $f(x) = a(x - (-1))(x - 3) = a(x + 1)(x - 3)$. The vertex of the parabola is given as $(1, -4)$. This means that when $x = 1$, $f(x) = -4$. Substitute the vertex coordinates into the factored form of the function: $-4 = a(1 + 1)(1 - 3)$ $-4 = a(2)(-2)$ $-4 = -4a$ Divide both sides by -4 to solve for a: $a = \frac{-4}{-4}$ $a = 1$ Therefore, the v.

26 [Foundational] • Target: Piecewise functions

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. What is the value of $f(0)$?

a. -3

b. -1

c. 1

d. 3 ✓

To find the value of $f(0)$, we substitute $x = 0$ into the function's equation: $f(x) = x^2 - 4x + 3$. Therefore, $f(0) = (0)^2 - 4(0) + 3 = 0 - 0 + 3 = 3$. This corresponds to the y-intercept of the graph.

27 [Foundational] • Target: Reading function graphs

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. What is the value of $f(1)$?

a. 0 ✓

b. -1

c. 2

d. 3

To find the value of $f(1)$, we substitute $x = 1$ into the function $f(x) = x^2 - 4x + 3$: $f(1) = (1)^2 - 4(1) + 3$ $f(1) = 1 - 4 + 3$ $f(1) = 0$

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28 [Foundational] • Target: Even, odd, and other function properties

Consider the function $f(x)$ defined by $f(x) = |x - 2| + |x + 1|$. Find the minimum value of $f(x)$ and the interval over which this minimum is attained.

a. **Minimum value 3 on the interval** $[-1, 2]$ ✓

b. Minimum value 2 on the interval $[-1, 2]$

c. Minimum value 3 on the interval $(-\infty, -1]$

d. Minimum value 2 on the interval $[2, \infty)$

The function is defined as $f(x) = |x - 2| + |x + 1|$. We can analyze this function by considering the critical points where the expressions inside the absolute values change sign, which are $x = 2$ and $x = -1$. This divides the number line into three intervals: Case 1: $x2$ In this interval, $(x - 2)$ is positive and $(x + 1)$ is positive. So, $f(x) = (x - 2) + (x + 1) = x - 2 + x + 1 = 2x - 1$. This is an increasing linear function. Combining these cases, we see that: - For $x2$, $f(x)$ increases. Therefore, the minimum value of.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: modeling a car rental cost

The graph of function g is obtained by shifting the graph of f 6 units to the right and 6 units down. Which of the following defines $g(x)$?

a. $g(x) = f(x - 6) - 6$ ✓

b. $g(x) = f(x - 6) + 6$

c. $g(x) = f(x + 6) - 6$

d. $g(x) = f(x + 6) + 6$

Horizontal translations: Shifting a graph c units to the right replaces x with $(x - c)$, while shifting c units to the left replaces x with $(x + c)$. Vertical translations: Shifting d units up adds d to the function, while shifting down subtracts d. Therefore, shifting 6 units to the right and 6 units down yields: $g(x) = f(x - 6) - 6$.

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30 [Intermediate] • Target: Evaluating functions

The graph of function g is obtained by shifting the graph of f 2 units to the left and 5 units down. Which of the following defines $g(x)$?

a. $g(x) = f(x + 2) - 5$ ✓

b. $g(x) = f(x - 2) - 5$

c. $g(x) = f(x + 2) + 5$

d. $g(x) = f(x - 2) + 5$

Horizontal translations: Shifting a graph c units to the right replaces x with $(x - c)$, while shifting c units to the left replaces x with $(x + c)$. Vertical translations: Shifting d units up adds d to the function, while shifting down subtracts d. Therefore, shifting 2 units to the left and 5 units down yields: $g(x) = f(x + 2) - 5$.

31 [Intermediate] • Target: Domain and range

The graph of function g is obtained by shifting the graph of f 4 units to the right and 7 units up. Which of the following defines $g(x)$?

a. $g(x) = f(x + 4) - 7$

b. $g(x) = f(x - 4) - 7$

c. $g(x) = f(x - 4) + 7$ ✓

d. $g(x) = f(x + 4) + 7$

Horizontal translations: Shifting a graph c units to the right replaces x with $(x - c)$, while shifting c units to the left replaces x with $(x + c)$. Vertical translations: Shifting d units up adds d to the function, while shifting down subtracts d. Therefore, shifting 4 units to the right and 7 units up yields: $g(x) = f(x - 4) + 7$.

32 [Intermediate] • Target: Composition of functions

The graph of function g is obtained by shifting the graph of f 4 units to the left and 7 units up. Which of the following defines $g(x)$?

a. $g(x) = f(x + 4) - 7$

b. $g(x) = f(x - 4) + 7$

c. $g(x) = f(x - 4) - 7$

d. $g(x) = f(x + 4) + 7$ ✓

Horizontal translations: Shifting a graph c units to the right replaces x with $(x - c)$, while shifting c units to the left replaces x with $(x + c)$. Vertical translations: Shifting d units up adds d to the function, while shifting down subtracts d. Therefore, shifting 4 units to the left and 7 units up yields: $g(x) = f(x + 4) + 7$.

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33 [Intermediate] • Target: Inverse functions

What is the maximum value of the function $f(x) = -x^2 + 10x - 20$?

a. 9

b. -5

c. 1

d. 5 ✓

The vertex of a parabola $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$: $x = -\frac{10}{2(-1)} = 5$ Substitute $x = 5$ into $f(x)$ to obtain the maximum value: $f(5) = -1(5)^2 + (10)(5) + (-20) = 5$ Alternatively, completing the square gives $f(x) = -1(x - 5)^2 + 5$, which clearly shows the vertex at $(5, 5)$.

34 [Intermediate] • Target: Transformations of graphs

What is the minimum value of the function $f(x) = 2x^2 - 20x + 53$?

a. 5

b. 7

c. 3 ✓

d. -5

The vertex of a parabola $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$: $x = -\frac{-20}{2(2)} = 5$ Substitute $x = 5$ into $f(x)$ to obtain the minimum value: $f(5) = 2(5)^2 + (-20)(5) + (53) = 3$ Alternatively, completing the square gives $f(x) = 2(x - 5)^2 + 3$, which clearly shows the vertex at $(5, 3)$.

35 [Intermediate] • Target: Piecewise functions

The graph of $f(x)$ is shifted 2 units left and 3 units up. Which expression gives the new function?

a. $f(x - 2) + 3$

b. $f(x + 2) + 3$ ✓

c. $f(x - 2) - 3$

d. $f(x + 2) - 3$

Left $\Rightarrow x + 2$ inside; up $\Rightarrow +3$ outside.

36 [Intermediate] • Target: Reading function graphs

What value of x is excluded from the domain of $f(x) = \frac{1}{x+3}$?

a. 3

b. **-3** ✓

c. -2

d. -1

$x + 3 = 0 \Rightarrow x = -3$ is excluded

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37 [Intermediate] • Target: Even, odd, and other function properties

The diagram shows the graphs of the exponential function $y = a \cdot b^x$ and the linear function $y = mx + c$. The exponential function passes through the points $(0, 5)$ and $(2, 45)$. The linear function passes through the points $(0, 5)$ and $(3, 20)$. Find the value of a.

a. 1

b. **5** ✓

c. 3

d. 9

For the exponential function $y = a \cdot b^x$: 1. Using the point $(0, 5)$: $5 = a \cdot b^0$ Since $b^0 = 1$ for any non-zero b, we have $5 = a \cdot 1$, which means $a = 5$. 2. Using the point $(2, 45)$ and the value of $a = 5$: $45 = 5 \cdot b^2$ Divide by 5: $9 = b^2$ Taking the positive square root (as bases of exponential functions are typically positive), $b = 3$. So, the exponential function is $y = 5 \cdot 3^x$. For the linear function $y = mx + c$: 1. Using the point $(0, 5)$: $5 = m(0) + c$ This gives.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: modeling a car rental cost • Cross-Lesson: 7.01 Trigonometry & Additional Topics

In the xy-plane, the perpendicular lines T and K intersect at the point (3,2). If line T contains the point (2,5), which of the following points is on line K?

a. (5,3)

b. (-2,1)

c. (1,2)

d. **(6,3)** ✓

Slope of T (through $(3, 2)$ and $(2, 5)$) $= -3$, so slope of $K = \frac{1}{3}$. Line $K: y = \frac{1}{3}x + 1$, which passes through $(6, 3)$.

39 [Challenging] • Target: Evaluating functions • Cross-Lesson: 8.01 Exponentials & Logarithms

What is the y-intercept of the line passing through points (5;5) and (-5;-1)?

a. -3

b. **2** ✓

c. 0

d. 4

Slope $= \frac{5-(-1)}{5-(-5)} = 0.6$. At $x = 0$: $y = 5 - 0.6(5) = 2$.

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40 [Challenging] • Target: Domain and range • Cross-Lesson: 1.01 Heart of Algebra

Consider the functions $f(x) = 2x + 3$ and $g(x) = x + 7$. At what x-value do the graphs of $f(x)$ and $g(x)$ intersect?

a. **4** ✓

b. 7

c. 3

d. 10

To find the x-value where the graphs of $f(x)$ and $g(x)$ intersect, we set the two functions equal to each other: $f(x) = g(x)$. Substituting the given expressions for $f(x)$ and $g(x)$: $2x + 3 = x + 7$ To solve for x, subtract x from both sides of the equation: $2x - x + 3 = x - x + 7$ This simplifies to: $x + 3 = 7$ Next, subtract 3 from both sides of the equation: $x + 3 - 3 = 7 - 3$ Which gives the solution: $x = 4$ Therefore, the graphs of $f(x)$ and $g(x)$ intersect at $x = 4$.

41 [Challenging] • Target: Composition of functions • Cross-Lesson: 2.01 Quadratics & Polynomials

If $f(x) = \frac{6x - 4}{2}$, which of the following represents $f^{-1}(x)$?

a. $\frac{2x + 4}{6}$ ✓

b. $\frac{6}{2x - 4}$

c. $\frac{2x - 4}{6}$

d. $\frac{6x + 4}{2}$

To find the inverse function $f^{-1}(x)$, set $y = f(x)$ and solve for x in terms of y: $y = \frac{6x - 4}{2}$ Multiply both sides by 2: $2y = 6x - 4$ Add 4 to both sides: $2y + 4 = 6x$ Divide by 6: $x = \frac{2y + 4}{6}$ Swapping variables gives $f^{-1}(x) = \frac{2x + 4}{6}$.

42 [Challenging] • Target: Inverse functions • Cross-Lesson: 4.01 Ratios, Rates & Data

The functions f and g are defined by $f(x) = 2x - 3$ and $g(x) = 4x^2 + 4$. What is the value of $f(g(4))$?

a. 104

b. 137

c. 133 ✓

d. 123

First evaluate the inner function $g(4)$: $g(4) = 4(4)^2 + 4 = 4(16) + 4 = 68$ Now evaluate f at 68: $f(68) = 2(68) - 3 = 136 - 3 = 133$.

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43 [Challenging] • Target: Transformations of graphs • Cross-Lesson: 5.01 Statistics & Probability

At what x-value do $f(x) = 1x + 8$ and $g(x) = 3x - 2$ intersect?

a. -5

b. **5 ✓**

c. 13

d. 6

Applying Functions & Graphs principles, $1x + 8 = 3x - 2 \Rightarrow x = 5$ Therefore, the verified correct option is 5.

44 [Challenging] • Target: Piecewise functions • Cross-Lesson: 6.01 Geometry

The graph of $y = (x - 9)^2$ is the graph of $y = x^2$ shifted how?

a. 9 units up

b. 9 units left

c. 10 units right

d. **9 units right ✓**

$y = (x - h)^2$ shifts right by h; here $h = 9 \Rightarrow$ shift right by 9

45 [Challenging] • Target: Reading function graphs • Cross-Lesson: 7.01 Trigonometry & Additional Topics

The table shows a company's annual revenue (in 1000s) for two years. Assuming the relationship is linear, what is the rate of change in revenue per year between 2013 and 2019?

a. 15,000/yr

b. 16,000/yr

c. **18,000/yr ✓**

d. 20,000/yr

$\frac{288 - 180}{2019 - 2013} = \frac{108}{6} = 18$, i.e. \18,000/yr.

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